Math 534 H - Spring 2026 Lecture 1

Mathematical Models $\longrightarrow$ PDE

- based general laws for ex. coming from quantum mechanics (mass, energy, momentum, etc).

or on experiments in nature which depend strongly on the features of the phenomena being under consideration. (Fick's law for diffussion, Fourier Law of heat conduction, etc)

A PDE on $\mathbb{R}^d$ is a relation among independent variables $x_1, x_2, x_3, \ldots x_d$, the dependente variable

$u = u(x_1, \dots, x_d)$ and the partial derivatives of u $\left(\frac{\partial u}{\partial x_j}, \frac{\partial u}{\partial x_i \partial x_j}, \dots \right)$

$$\uparrow \qquad \uparrow$$

$$u_{x_j} \qquad u_{x_i x_j}$$

(*)

$$\boxed{F(x_1, x_2, \dots x_d, u, u_{x_1}, \dots, u_{x_d}, u_{x_1 x_1}, u_{x_1 x_2}, \dots) = 0}$$

u is the unknown and is a function of d-variables

Def.: The order of the PDE is the highest order of differentiation occuring in (*)

PDE :
- → LINEAR PDE : when F is linear w.r.t u and all its partial derivatives
- → NONLINEAR PDE

For nonlinear PDE we can distinguish certain types depending on the nonlinearity:

- <u>SEMILINEAR EQUATIONS</u>: F is nonlinear only on u but it is linear w.r.t all the derivatives of u that appear and has coefficients which at most depend only on $\mathbf{x} = (x_1, \dots, x_d)$

- <u>QUASILINEAR EQUATIONS</u>: F is linear w.r.t the highest order derivative of u that appears with coefficients depending at most only on $\mathbf{x}$, u, and lower order derivatives of u.

Fully Nonlinear Equations: F is nonlinear w.r.t to the highest order derivatives of u that appear.

Examples of Linear Equations

① **Transport Equations** (with constant coefficients)

$$u_t + \mathbf{v} \cdot \nabla_x u = 0 \qquad \vec{\mathbf{v}} = \vec{\mathbf{v}}(x) \text{ given}$$

(first order)

$$\mathbb{R}^{d+1} = (\underbrace{x_1, \dots, x_d}_{\mathbb{R}^d}, \underbrace{t}_{\mathbb{R}})$$

Describes the transport of a solid substance (pollutant, dye, ..) along a channel with stream speed $\vec{\mathbf{v}}$. Here $u = u(x,t)$ measures the

concentration of the substance.

(2) <u>Wave equation</u> (second order)

$$u_{tt} - c^2 \Delta_x u = 0 \qquad c \text{ constant}$$

$$\Delta u = u_{x_1 x_1} + u_{x_2 x_2} + \dots + u_{x_d x_d}$$

(Laplacian of u)

Describes the propagation of for example, transversal waves of small amplitude in a perfectly elastic chord (violin) in 1D or a membrane (drum) in 2D. In 3D it governs the propagation of electromagnetic waves in a vacuum, or of small

amplitude sound waves.

Here $u = u(x, t)$ may represent the amplitude

$x \in \mathbb{R}^d$, $t \in \mathbb{R}$ and c = propagation speed

③ <u>Diffussion or Heat Equation</u> (second order)

$$u_t - k \Delta_x u = 0 \qquad \boxed{k>0}$$

For ex. it describes the conduction of heat through a homogeneous and isotropic medium (air; water; glass).

$u = u(x,t)$ is the temperature and <u>$k>0$</u> encodes the thermal properties of the medium.

④ <u>Laplace's Equation</u> (second order)

$$\Delta_x u = 0 \qquad u = u(x) \quad x \in \mathbb{R}^d$$

The wave eq. or diffusion eq. model evolution (time dependent) phenomena. The Laplace equation describes the corresponding <u>steady-states</u> in which the solution doesn't depend on time any more.

We also have the <u>inhomogeneous version</u>

$$\Delta_x u = f \quad \text{where } f = f(x) \text{ <u>is given</u>}$$

↳ forcing term.

Called the POISSON equation.

⑤ Black-Scholes Equation (second order)

$x \geq 0$, $t \geq 0$ $\quad u = u(x,t)$

$$u_t + \underbrace{\frac{1}{2}\sigma^2 x^2}_{\text{coefficient}} u_{xx} + \underbrace{r x}_{\text{coefficient}} u_x - \underbrace{r}_{\text{constant coefficient}} u = 0$$

(σ, r are given constants)

σ → volatility

r → rate

⑥ Vibrating plate (fourth order)

$$u_{tt} - \Delta^2 u = 0 \qquad x \in \mathbb{R}^2$$

$$\Delta^2 u = \Delta(\Delta u) = \frac{\partial^4 u}{\partial x_1^4} + 2\frac{\partial^4 u}{\partial x_1^2 \partial x_2^2} + \frac{\partial^4 u}{\partial x_2^4}$$

(BILAPLACEAN or BIHARMONIC operator)

7 Schrödinger equation (second order)

(linear)

$$-i\, u_t = \Delta_x u + V(x) \cdot u$$

i = complex unit

$i = \sqrt{-1}$

$V = V(x)$ given potential (function).

$u = u(x,t)$ is complex-valued

This eq. is fundamental in quantum mechanics; it governs the evolution of a particle subject to a potential V.

The function $|u|^2$ = probability density.